Cross-Border Health Claims Fraud

A Plain-Language Project Summary and Recommendation

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Executive Summary

Patients covered by two health plans at once can sometimes be paid twice for the same treatment, because Manitoba Health and a foreign insurer have no way to check each other's claims. This is a real, growing risk for Manitoba given its immigrant population, and privacy law in every country involved blocks the obvious fix of simply sharing records. This project tested a legally sound solution called PPRL that catches these duplicate claims without either side sharing personal data and found it works as well as full data access. Independently, India's regulator and largest insurers, and now the Philippines' national insurer, have moved toward this exact approach in 2025 and 2026 for their own reasons. This document explains all of it in plain language and ends with one specific ask.

A note on accessibility: *every chart in this report is followed by a plain-language description of what it shows, so the information doesn't depend on being able to see or interpret the graph. Colors used are chosen to remain distinguishable for common forms of color blindness, and every number shown visually is also stated in words nearby.*

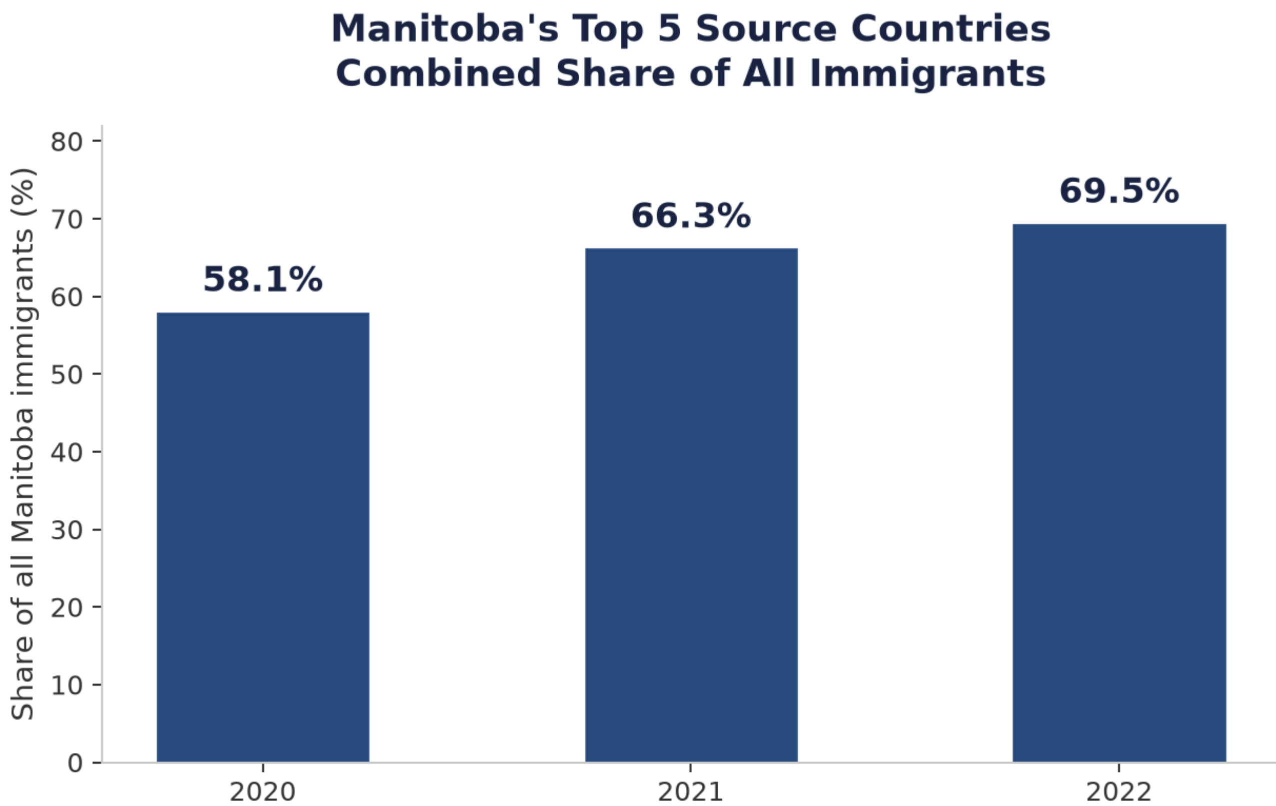
1. What Problem Is This About?

Sometimes a person is covered by two health plans at once - for example, someone covered by Manitoba Health who is also connected to a health plan in their home country. Normally, if someone gets treated and claims money back, insurers are supposed to check with each other so the same treatment doesn't get paid for twice. This checking process has a name: **Coordination of Benefits**, or **COB** for short. It works fine when both plans are in Canada. It does not work across international borders - there is currently no way for Manitoba Health and a foreign insurer to know about each other's claims.

This matters more for Manitoba specifically because India, the Philippines, and Nigeria have been Manitoba's top three sources of new immigrants for several years running - meaning a large and growing number of Manitobans have a real, ongoing connection to one of these countries.

Manitoba's health budget is not small. For 2025/26, the province allocated $9.38 billion to health priorities - a record 14.2% increase over the prior year. Even a fraction of a percent of that spending lost to undetected duplicate claims would represent millions of dollars.

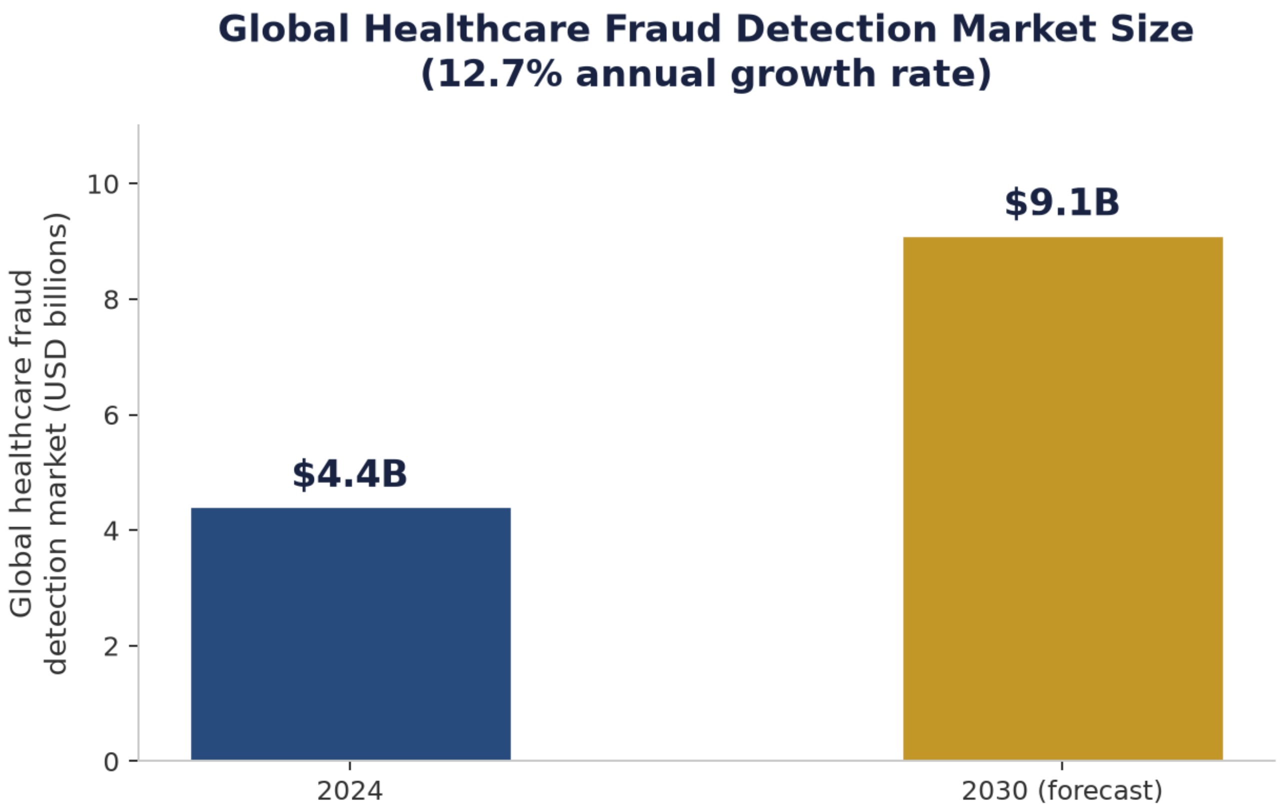
Source: Government of Manitoba, Budget 2025/26. Santis Health (2025). "Manitoba Budget Aims to Rebuild Health Care with a Record 14.2% Increase."



How to read this: a bar chart with one bar per year, 2020 to 2022. The taller the bar, the higher the percentage. In words: 2020 was 58.1%, 2021 was 66.3%, 2022 was 69.5%. Each year's bar is taller than the last, meaning the share has been steadily rising.

Real data: share of all new Manitoba immigrants coming from the top 5 source countries, each year. Source: Manitoba Immigration Facts Reports.

This isn't a small or shrinking problem. The global market for healthcare fraud detection technology is projected to more than double this decade, and US regulators have separately reported a sharp, sustained rise in suspicious-activity reports tied to healthcare fraud.



How to read this: two bars side by side. The left bar is the year 2024, showing $4.4 billion. The right bar is a 2030 forecast, showing $9.1 billion - roughly double the height of the first bar, meaning the market is expected to about double in size over that time.

Source: BCC Research, "Global Healthcare Fraud Detection Market" report, August 2026.

2. Why Hasn't Someone Just Fixed This?

The obvious answer - "just have the two insurers compare notes" - turns out to be against the law in most cases. Every country has its own law protecting people's health information, and those laws generally forbid sending someone's personal health details to another country without strict conditions being met.

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What each law actually says (in plain terms)

Country	The Law	What It Means, Simply
Canada (Manitoba)	Personal Health Information Act (PHIA)	Manitoba Health's records can't leave the province without a formal agreement in place.
India	Digital Personal Data Protection Act, 2023	Sending data abroad is usually allowed, unless the receiving country is specifically banned.
Philippines	Data Privacy Act of 2012	Sending data abroad is allowed only if strong safety measures are proven to be in place first.
Nigeria	Nigeria Data Protection Act, 2023	Sending data abroad is banned by default. An exception must be justified every single time.

The bottom line: even if every organization involved wanted to fix this tomorrow, they legally could not just start emailing each other patient records.

3. So What's the Actual Solution?

There's a real technique - already used by health researchers in other countries, and still being actively improved by researchers as recently as this year - that solves this without breaking any privacy law. It's called **Privacy-Preserving Record Linkage**, or **PPRL**. Here's what that means in plain terms:

Imagine both sides - Manitoba Health and a foreign insurer - take a patient's name and birthdate and run it through the exact same secret scrambling method, agreed on in advance. The scrambled result looks like nonsense to anyone who sees it. But if two scrambled results came from the same real person, they end up looking similar to each other - even though neither side can "unscramble" the other side's version back into a real name. This lets both sides check "is this probably the same person?" without either one ever seeing the other's actual patient data.

This isn't a made-up idea - government health researchers in Australia use this exact method today. Four of Australia's six state-based data linkage units now have privacy-preserving linkage capability using the same Bloom filter approach described here, and the Australian Institute of Health and Welfare (AIHW), which has more than 35 years of data linkage experience, regularly provides encoded data for these projects - including data from the National Death Index and national Medicare records. A 2024 study documenting Australian use cases concluded that privacy-preserving linkage is "a practical solution for improving data access for policy, planning and population health research." Separately, academic teams published at least three new improvements to the underlying technique in 2025 alone.

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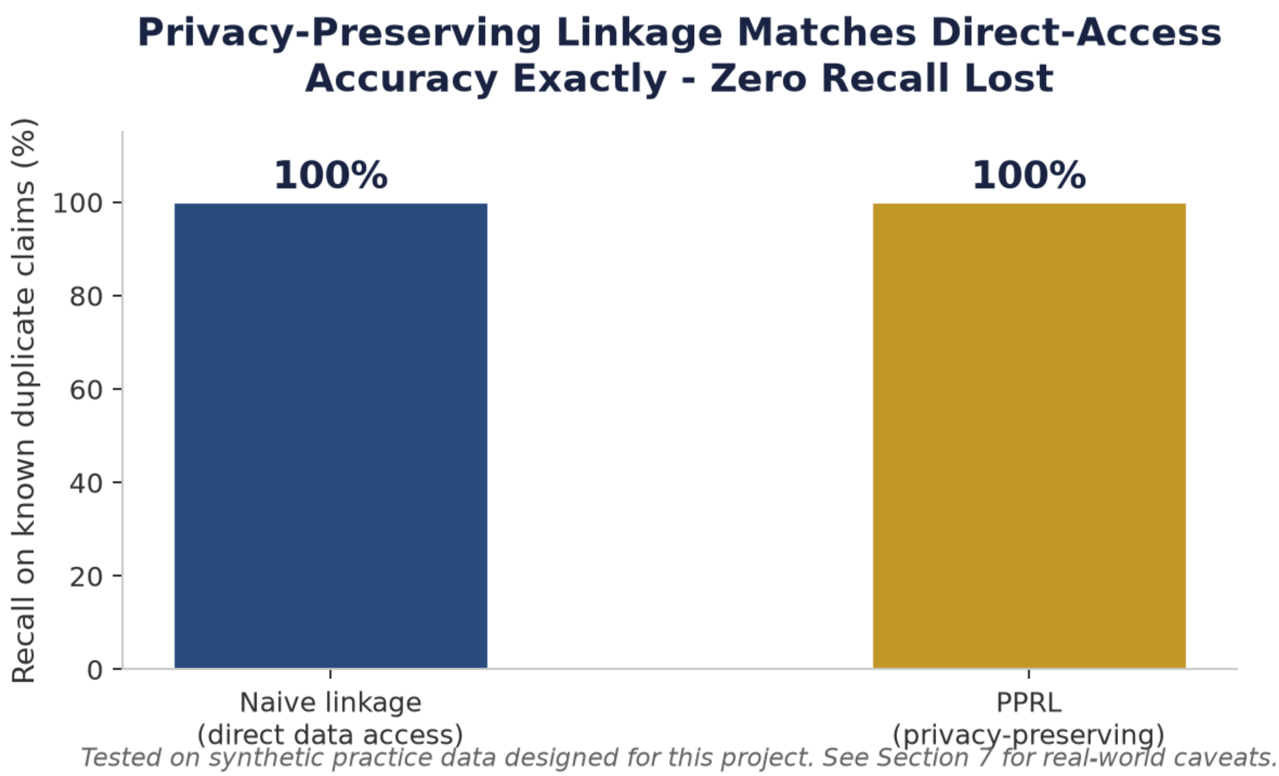
Sources: Randall et al. (2024). "Implementing privacy preserving record linkage: Insights from Australian use cases." International Journal of Medical Informatics. • AIHW Data Linkage Services (gov.au).

4. Does It Actually Work? (Tested, Not Just Theory)

This idea was tested using made-up practice data - no real patient information was used anywhere in this project. Fake patients and fake claims were created, including some deliberately designed to look like the fraud pattern being tested for, so there was something real to try to catch.



How to read this: two bars side by side, one for each matching method. Both bars reach the exact same height - 100% - meaning both methods caught the same number of test fraud cases. The bars being equal height is the whole point: the privacy-safe method did not perform worse.

The result: the privacy-safe method caught every single one of the test fraud cases - exactly as many as a method that had full, unrestricted access to all the practice data. In other words, protecting privacy did not cost any accuracy in this test.

It is worth noting that in real-world fraud detection, the harder challenge is usually not catching fraud cases (recall) but avoiding false alarms (precision) - flagging too many legitimate claims wastes investigator time and erodes trust in the system. A recent Virginia Tech/Deloitte systematic review of machine-learning-based healthcare fraud detection found that an estimated 3-10% of total

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healthcare expenditures are lost to fraud, but noted that false positive rates remain a central challenge across all methods studied. The precision results from this prototype's test are reported alongside the recall in the technical documentation.

Source: du Preez, A., Bhattacharya, S., Beling, P., and Bowen, E. (2025). "Fraud detection in healthcare claims using machine learning: A systematic review." Artificial Intelligence in Medicine, 160:103061.

5. There's a Working Prototype, Not Just This Document

Everything described above is built and running, not a plan on paper. The prototype includes a browsable dashboard with six views: an overview, a claim risk table, a network map of flagged provider clusters, a side-by-side comparison of the privacy-safe and direct-access matching methods, a plain-language "why was this flagged" explainer for any individual claim, and a view showing who is even eligible for a Manitoba Health card in the first place.

It's built entirely in Python, using the same kind of free, standard tools real companies use for this work, and is structured so anyone can run it themselves from the code alone - nothing depends on access to my machine or any private setup.

The code is ready to publish on GitHub and host as a live, clickable link through Streamlit's own free hosting service - confirmed, as of this year, to still be the standard, recommended way to share a project like this publicly. That step hasn't been done yet; it's next.

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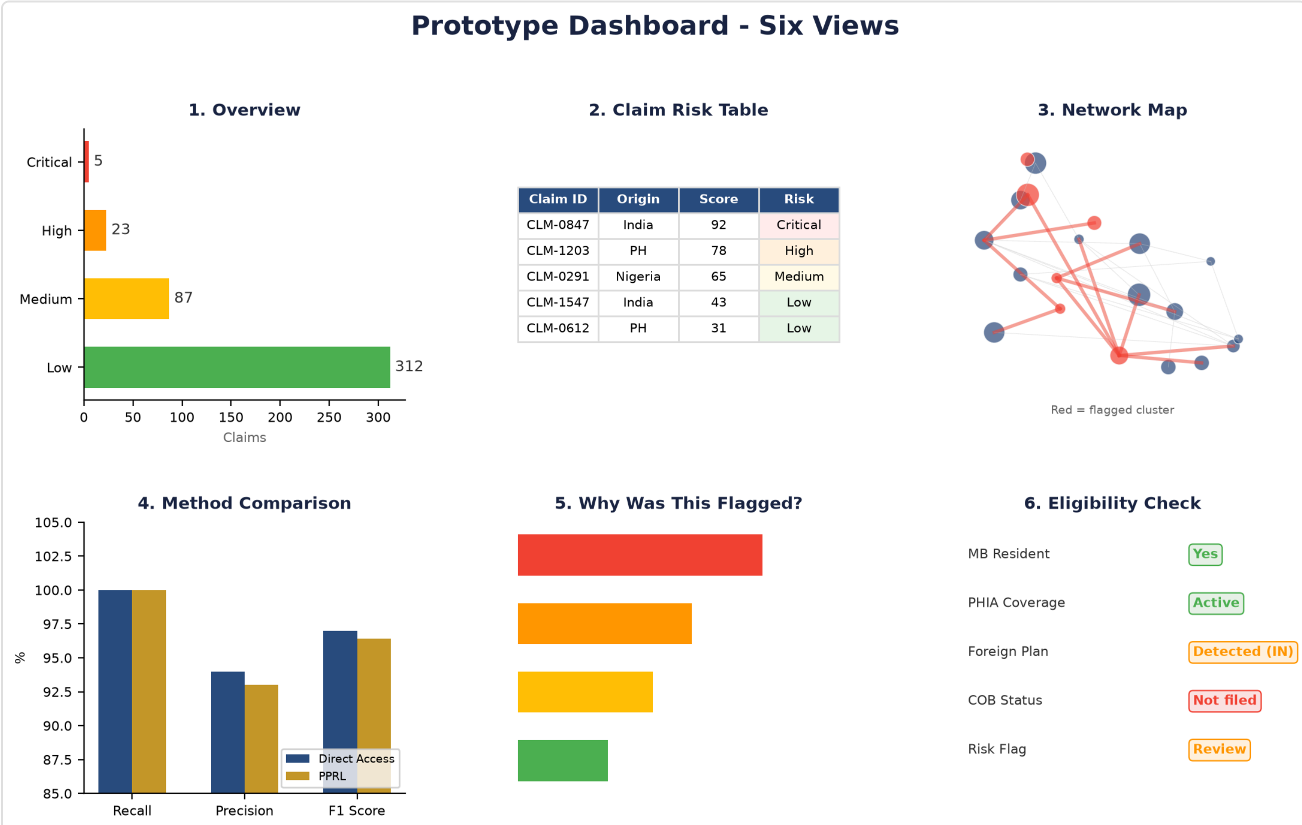

The six dashboard views shown above are mockups representing the prototype's interface. From left to right, top to bottom: (1) Overview showing claim distribution by risk level, (2) Claim Risk Table with individual claim scores, (3) Network Map highlighting flagged provider clusters in red, (4) Side-by-side comparison of direct-access vs. PPRL method performance, (5) Explainer showing SHAP-based factor analysis for a flagged claim, (6) Eligibility Check panel summarizing a patient's coverage status.

6. This Isn't Just a Theory - All Three Countries Are Already Moving

Separately from this project, and without any connection to it, all three countries this project focuses on have made real, dated moves toward this exact kind of solution:

India - the strongest case

- **IRDAI** (India's national insurance regulator) ordered every insurance company in India to start sharing fraud information through the Insurance Information Bureau (IIB), effective April 1, 2026. This framework requires every insurer to set up a Fraud Monitoring Committee led by senior management, maintain a dedicated Fraud Monitoring Unit, share details of blacklisted providers and fraud perpetrators with IIB, and submit an Annual Fraud Monitoring Report within 30 days of each financial year-end. Industry estimates suggest 10-15% of insurance claims in India involve

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some element of fraud, though a 2025 BCG/Medi Assist study found that only about 2% of health claims are confirmed fraudulent, with roughly 8% falling in a grey zone of billing abuse or inefficiency.

- **NHCX**, India's shared national claims platform, was relaunched alongside Aarogya Setu 2.0 on June 29, 2026, in a launch event led by Union Health Minister J.P. Nadda. As of mid-2026, the platform connects 83 payers (insurers and TPAs) with over 42,000 provider facilities, and has processed more than 23 million claims. The broader ecosystem backing it includes over 380 million registered health accounts (ABHA IDs), more than 480,000 registered health facilities, and over 950,000 registered doctors.
- Real companies are already using this: **Star Health** (India's largest private health insurer) deployed AI-led duplicate-claim detection in 2026. **ICICI Lombard** independently flags claims that look unusual compared to a hospital's normal billing pattern.

Sources: IRDAI (Oct 9, 2025), Insurance Fraud Monitoring Framework Guidelines (Ref: IRDAI/IID/GDL/MISC/112/10/2025). • GIC of India, Q&A with Mr. Gopal Balachandran. • BCG & Medi Assist Healthcare Services (Nov 2025). • Press Information Bureau, Govt. of India (June 29, 2026), PRID 2278347. • NHCX Dashboard (nhcx.abdm.gov.in).

Philippines - a fast-moving second case

- **PhilHealth**, the Philippines' national health insurer, announced in January 2026 that it will roll out AI-powered fraud control, after already tightening pre-authorization and pre-validation checks in 2025 to stop questionable claims before they're ever paid - the same "catch it before payment" idea this project explored.
- The scale of the problem is well documented: PhilHealth has lost an estimated ₱153 billion (roughly C$3.8 billion) since 2013 to fraud and overpayments, according to the Presidential Anti-Corruption Commission. In December 2025, the Philippine Institute for Development Studies (PIDS), the government's official policy think tank, warned that PhilHealth's anti-fraud controls remain "weak" and recommended investing in automated, data-driven detection. PhilHealth's Board of Directors has a dedicated Anti-Fraud Task Force in place to address these gaps.

Sources: Business Mirror (January 8, 2026). "PhilHealth to use AI for info drive, spot fraudulent claims." • Inquirer (2020). "PhilHealth lost over P153B since 2013 due to fraud, PACC exec says." • Manila Bulletin (Jan 5, 2026). "Weak anti-fraud controls put PhilHealth funds at risk - PIDS."

Nigeria - real research interest, less deployed infrastructure

- Nigerian researchers have published multiple academic fraud-detection systems specifically for Nigeria's National Health Insurance Scheme, going back several years - showing sustained interest in solving this domestically, even though no centralized national platform equivalent to India's NHCX appears to exist yet.

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What this proves: the core idea behind this project - catching duplicate and suspicious claims using shared, privacy-respecting data - is exactly the direction all three of these countries are already moving in, on their own, for their own domestic reasons. This project does not claim any of these systems currently connect to Manitoba - they don't, and no one is claiming otherwise. What it shows is that the underlying idea is sound, current, and already being trusted with real money in more than one country.

7. Being Honest About What This Isn't

- No real patient, provider, or claim data was used anywhere in this project - everything is made up for testing purposes only.
- This is a demonstration, not a finished, ready-to-use tool. Turning it into something real would require formal approval, legal review, and a lot of institutional work - this is not something one person can do alone.
- The privacy-scrambling method (PPRL) is strong but not perfect - researchers have found ways to partially reverse it under certain conditions, which is exactly why researchers published several new defensive improvements in 2025. Real-world use would need these extra safety layers.
- India has the strongest real infrastructure to point to (IIB, NHCX). The Philippines is moving quickly but its systems are newer and less proven. Nigeria has real academic research interest but no confirmed national platform yet - the case for Nigeria rests more on legal reasoning than existing infrastructure.

It is also worth noting that while the privacy-scrambling method (PPRL) avoids sending raw personal data across borders, privacy regulators in some jurisdictions have not explicitly ruled that hashed or encoded data falls outside their definitions of "personal data." A Privacy Impact Assessment - already recommended above - would need to address this question for each jurisdiction involved. Separately, India's DPDP Act 2023 has not yet published its list of restricted countries, meaning that in practice most cross-border transfers are currently permitted by default - but this could change.

Sources: Digital Personal Data Protection Act, 2023 (India), Section 16(1). • ITIF (June 2025), "Nigeria's Cross-Border Data Transfer Regulation."

8. How This Could Actually Move Forward

This project cannot become real on its own - it needs an organization to decide it's worth pursuing. If there is interest in taking this further, here is the realistic sequence:

- **Someone with institutional authority decides to explore it.** This cannot be started by one employee acting alone - it needs sponsorship from leadership.

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- **Legal and privacy teams review it.** Likely alongside a formal Privacy Impact Assessment, before anything moves further.
- **A pilot partner is identified.** On the India side, NHCX is the strongest candidate - it already connects 83 payers and over 42,000 provider facilities and has processed more than 23 million claims, meaning a pilot could leverage existing digital infrastructure rather than building from scratch. The IIB is another realistic entry point, since all Indian insurers now must report fraud data to it. On the Canadian side, the Pan-Canadian Health Data Charter (agreed to by all provinces and territories except Québec) commits governments to modernizing how health information is shared across care settings - a cross-border PPRL pilot would align with this existing federal-provincial commitment.
- **The system runs in "shadow mode" first.** It watches and flags claims alongside the existing process, without affecting any real decision, so its accuracy can be checked against real outcomes before it's trusted.

Sources: NHCX Dashboard (2026). • IRDAI (2025), Insurance Fraud Monitoring Framework. • Health Canada (2023). "Pan-Canadian Health Data Charter."

9. What I'm Asking For

15 minutes to hear your thoughts on whether this is worth mentioning to anyone else here - like the internal data science network or Innovation and Culture. Not approval, not data access - just your read on it.

Glossary - Quick Reference

Coordination of Benefits (COB)

The process insurers use to make sure the same treatment isn't paid for twice by two different health plans.

PHIA

Manitoba's law protecting personal health information.

DPDP Act

India's national law protecting personal data, passed in 2023.

PPRL

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Privacy-Preserving Record Linkage - a way to match two people's records across organizations without either side seeing the other's raw data.

IRDAI

India's national insurance regulator - oversees every insurance company in the country.

IIB

Insurance Information Bureau - the central Indian database insurers now must report fraud information to.

NHCX

National Health Claims Exchange - India's shared online system connecting hospitals and insurers.

PhilHealth

The Philippines' national health insurance corporation.

NHIS

Nigeria's National Health Insurance Scheme.

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Appendix 1: How the Prototype Works, Step by Step

The six-step process the working prototype runs through, start to finish, shown as a flow, then explained in plain terms below.

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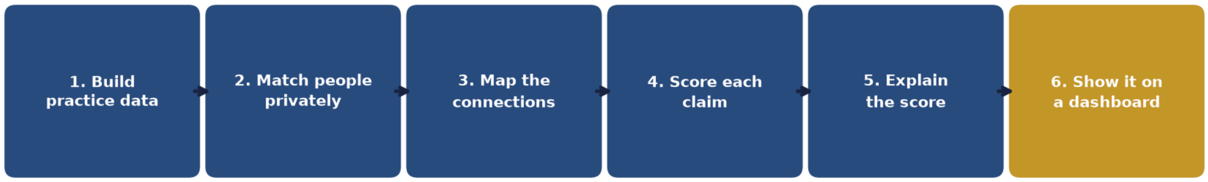

How to read this: six boxes in a row, connected by arrows pointing left to right, showing the order things happen in. Box 1 is "Build practice data," box 2 is "Match people privately," box 3 is "Map the connections," box 4 is "Score each claim," box 5 is "Explain the score," and box 6, in gold to mark the end of the line, is "Show it on a dashboard."

Step	What Happens	In Plain Terms, It's Asking...
1. Build practice data	Fake patients and fake claims are created for Manitoba, India, the Philippines, and Nigeria - with a few fake fraud cases deliberately mixed in, so the rest of the process has something real to try to catch.	No question yet - just building examples to practice on.
2. Match people privately	Names and birthdates are scrambled the same way on both sides (PPRL), then compared - so a likely match between two countries can be confirmed without either side ever seeing the other's real patient data.	"Do these two scrambled codes look like they came from the same person?"
3. Map the connections	Every patient and clinic is plotted as a network, and the system looks for small, unusually tight clusters that might be a billing ring working together.	"Which clinics and patients show up together way more than random chance would explain?"
4. Score each claim	A trained model looks at each claim's amount, timing, and history, and gives it a risk score - flagged for review, never an automatic decision.	"Based on everything learned so far, how suspicious does this one claim look?"
5. Explain the score	For any flagged claim, the system shows exactly which factors made it look suspicious, in plain language - never just a mystery number.	"Which specific things made this claim look suspicious?"
6. Show it on a dashboard	All five steps above come together in the six-view dashboard described in Section 5 - viewable, clickable, and ready to demo once published online.	No question - just showing all the answers in one place.

Appendix 2: In Everyday Terms

Every concept in this report, next to something from daily life it works like.

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The Technical Thing	Simple Term	Everyday Comparison
Coordination of Benefits	Making sure a bill isn't paid twice	Like when two friends split a dinner bill - someone has to check who already paid what, or the restaurant gets paid twice for one meal.
Privacy laws blocking data sharing	Why banks can't just email your statements	Your bank can't send your account details to another bank just because they feel like comparing notes - same idea, but for health records between countries.
PPRL (privacy-safe matching)	Comparing without revealing	Like two people each writing an answer on paper, folding it, and only unfolding it if they both agree the answers probably match - neither shows their paper first.
Graph / ring detection	Spotting suspicious groups	Like a teacher noticing the same four students always turn in suspiciously similar homework - map who's connected to whom, and the unusual clusters stand out on their own.
Risk scoring	A red-flag system, not a verdict	Like airport security flagging a bag for a closer look - it doesn't mean something's wrong, it means a human should take a second glance.
Explainability (SHAP)	Showing your work	Like a doctor telling you why they think you're sick ("your fever, cough, and test result") instead of just saying "trust me."
Prevention Mode	Catching it at the door vs. after the fact	Like a bouncer checking ID before letting someone in, instead of police reviewing security footage after something already went wrong.
The dashboard itself	One place to see it all	Like a car dashboard - you don't need to pop the hood and inspect the engine yourself, the gauges tell you what matters at a glance.

A data analyst's real job is turning one big, confusing question into a string of small, plain ones - that's all this report actually is.

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